Uniformity Trials

Studying soil heterogeneity: fertility maps, serial correlation, strip mean squares, Smith's law and optimum plot size.

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1 What is a uniformity trial?

A trial to study the **nature and extent of fertility variation** in a field.

- One crop variety over the whole field; all inputs applied **uniformly**.
- Remove a border, then harvest separately by **basic units** (smallest recorded unit).
- Yield differences among basic units measure **soil heterogeneity**.

Helps decide: plot size, plot shape, block size and blocking direction.

2 Fertility contour map

A map of yields over the field. **Moving averages** smooth random variation; areas of similar fertility are joined by contour lines.

Rice example: 12 m × 17 m field, 1 m border removed → 10 m × 15 m → 150 basic units of 1 m × 1 m.

3 Serial correlation

Neighbouring units

$$r_s = \frac{\sum X_i X_{i+1} - \frac{(\sum X_i)^2}{n}}{\sum X_i^2 - \frac{(\sum X_i)^2}{n}}$$

Computed horizontally and vertically. High r_s: neighbouring yields are similar. Low r_s: little linear association at that lag (not proof of patches).

4 Mean square between strips

Vertical strips

$$MS_V = \frac{\frac{\sum V_i^2}{r} - \frac{G^2}{n}}{c-1}$$

Horizontal strips

$$MS_H = \frac{\frac{\sum H_i^2}{c} - \frac{G^2}{n}}{r-1}$$

A simpler method with the same aim as serial correlation. The larger mean square shows the direction of greater variation. $n = r \times c$.

5 Fairfield Smith's variance law

Variance law

$$V_x = \frac{V_1}{x^b}$$

Log form (linear)

$$\log V_x = \log V_1 - b \log x$$

Regression through origin

$$Y = cX, \quad c = -b$$

V_x = variance of yield per basic unit for plots of x units; b = **Smith's index of soil heterogeneity**, idealised range 0 to 1. $Y = \log V_x - \log V_1$, $X = \log x$.
Example: $b = 0.3607$, adjusted ≈ 0.4116.

6 Simulated plots

Variance per unit area

$$V_x = \frac{Q_x}{rc - x}$$

Combine basic units into plots of x units ($w = rc/x$ plots, fitting the field exactly). A plot's yield is the **sum** of its units.

7 Maximum curvature method

Plot **plot size** (X-axis) against **coefficient of variation** (Y-axis). CV generally falls as plot size rises; the point of **maximum curvature** (the bend) is taken as the optimum plot size.

Larger plots
Lower CV, but fewer plots fit a fixed area.

Caution
The bend depends on axis scales and the available arrangements: treat it as a guide.

! Common mistakes

- × Different shapes partition the same field, so their variances are **dependent**: no ordinary F test.
- × High b alone doesn't prove a patch pattern.

? Fill in the blanks

1. If Smith's index b = 0, a larger plot _____ the variance per unit area. **[does not reduce]**
2. If b = 1, doubling the plot size _____ the variance per unit area. **[halves]**
3. The slope of log V_x plotted against log x equals _____. **[–b]**
4. A 10 m × 15 m net area cut into 1 m² units gives _____ basic units. **[150]**
5. In the maximum curvature method, CV is plotted against _____. **[plot size]**
6. In a uniformity trial every basic unit gets the _____ variety and management. **[same]**

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